RISK 201: Equivalence Equations

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2.1 Equations of value

Some observations:

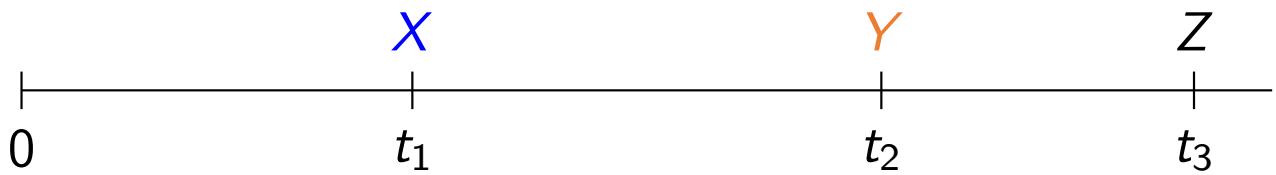
- Money has time value.
- We need **dates** attached to each cash flow to be able to make comparisons.
- If we are indifferent between two options, we say the cash flows are **equivalent**. Formally:

X due on a given date is equivalent
at a given compound interest rate i
to an amount Y due n periods later

if $Y = X(1+i)^n$

Transitivity under compound interest

At a given compound interest rate,
if X is equivalent to Y, and Y is equivalent to Z,
then X is equivalent to Z.



$$Y = X(1+i)^{t_2-t_1}$$
$$Z = Y(1+i)^{t_3-t_2}$$

$$\Rightarrow Z = X(1+i)^{t_2-t_1}(1+i)^{t_3-t_2} = X(1+i)^{t_3-t_1}$$

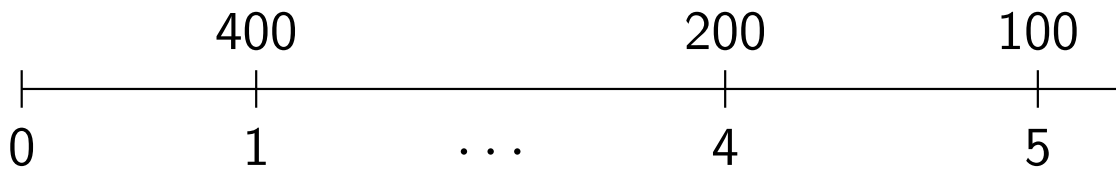
Sets of cash flows

Some more observations:

- Often, we care about a **set of dated cash flows** and want to know their combined value.
- Adding up values in the set makes no sense if they are at different dates.
- Solution: replace the dated cash flows in the set with equivalent cash flows all due on the **same** date, THEN add them up $\rightarrow$ **dated value of the set**
- The common date is called the **valuation date**, or **comparison date**, or **focal date**.

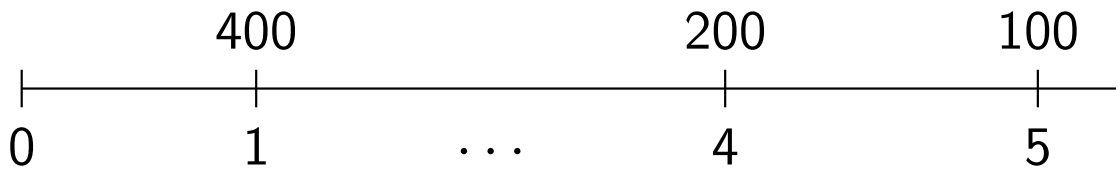
Example 2.1

An investment guarantees to pay \$400 in 1 year, \$200 in 4 years, and \$100 in 5 years. What is the value of this investment today if $i = 5\%$?



Example 2.1, continued

What is the value of this investment a year from now? Four years from now? 10 years from now?

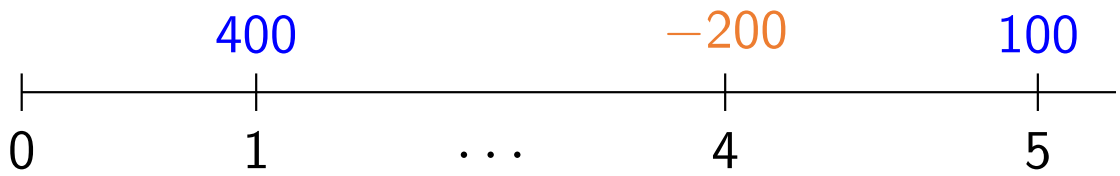


Keeping track of cash in and out

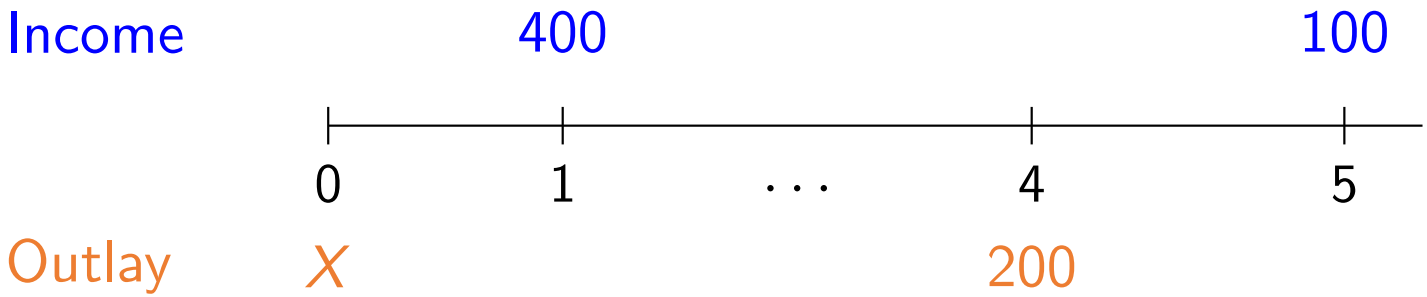
- Some of the cash flows in our set may be income, some may be outlays (aka outgo):
  - A loan vs the repayments
  - An initial investment vs its returns
- Attach opposite signs to cash income and outgo $\rightarrow$ dated **directional** cash flows

Example 2.2

An investment guarantees to pay \$400 in 1 year, and \$100 in 5 years, but requires an additional cash injection of \$200 after 4 years. What is the value of this investment today if $i = 5\%$?



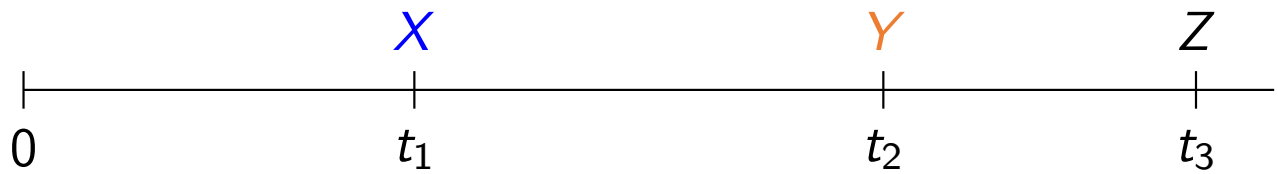
Example 2.2, continued



dated value of income = dated value of outlay
on a common valuation date, at a given rate i

Simple interest & simple discount

- **Only** compound interest has the transitive property for accumulation!
- Under simple interest (or simple discount), if X is equivalent to Y, and Y is equivalent to Z, then X is NOT equivalent to Z.



$$Y = X\big(1 + (t_2 - t_1)i\big), \qquad Z = Y\big(1 + (t_3 - t_2)i\big),$$

$$\begin{aligned} \Rightarrow Z &= X\big(1 + (t_2 - t_1)i\big)\big(1 + (t_3 - t_2)i\big) \\ &\neq X\big(1 + (t_3 - t_1)i\big)! \end{aligned}$$

Types of problems

Given	Find	Technique	Example
Two or more streams of payments, interest rate, times of payments	Preferred stream of payments	▶ Choose valn date ▶ Value all cash flows ▶ Compare	2.3
Amounts of some payments, interest rate, times of payments	Missing payment(s) that make cash flows equivalent	▶ Choose valn date ▶ Equate value of all cash flows at that date ▶ Solve for missing payment(s)	2.4

Example 2.3 (Brown & Kopp 1.10.A17)

A piece of land can be purchased by paying \$500,000 cash or \$200,000 down and equal payments of \$200,000 at the end of two and four years respectively. To pay cash, the buyer would have to withdraw money from an investment earning interest at $i^{(2)} = 8\%$. Which option is better and by how much?

Example 2.4 (Brown & Kopp 2.1.A15)

Mrs. Singh borrows \$3000, due with interest at $i^{(12)} = 9\%$ in 2 years. The lender agrees to let Mrs. Singh repay the loan with a payment of \$1000 in 6 months, \$1500 in 12 months, and \$$X$ in 30 months. If money is worth $i^{(2)} = 6\%$, what is the value of X?

2.2 Determining rate and time

Given	Find	Technique
Amounts of all payments, interest rate	Time of payment that makes cash flows equivalent	▶ Choose valn date ▶ Equate value of all cash flows at that date ▶ Solve for t
Amounts of all payments, times of payments	Interest rate that makes cash flows equivalent	▶ Choose valn date ▶ Equate value of all cash flows at that date ▶ Solve for i

Finding the interest rate

Algorithm

1. Consider the compounding period
2. Write the equation of value in terms of j, the effective rate per compounding period
3. Solve for j
4. Convert to whatever effective or nominal rate is required.

Finding the interest rate

On your financial calculator

1. N = the number of compounding periods
2. PV = − present value
3. FV = future value
4. PMT = periodic payments (zero in this context)
5. CPT I/Y (interest rate per compounding period)

NOTE: The rate will be given as a %, not a decimal

Example 2.5 (Brown & Kopp 2.2.A10)

If an investment grows by 50% in 4 years, what rate of interest $i^{(4)}$ is being earned?

Example 2.5, continued

On your financial calculator

1. N $= 16$
2. PV $= -100$
3. FV $= 150$
4. PMT $= 0$
5. CPT I/Y

Finding the time

Algorithm

1. Consider the compounding period
2. Write the equation of value in terms of j, the effective rate per compounding period (if annual, use i)
3. Solve for the time (in the exponent) by taking logs of both sides

Finding the time

On your financial calculator

1. PV $=-$ present value
2. FV $=$ future value
3. PMT $=$ periodic payments (zero)
4. I/Y $=$ interest rate per compounding period (%)
5. CPT N (the number of compounding periods)

Example 2.6

How long does an investment take to double at a given rate of compound interest?

i	t
2%	35.00
4%	17.67
6%	11.90
8%	9.00
10%	7.27

Rule of 72

On your financial calculator

1. PV $= -100$
2. FV $= 200$
3. PMT $= 0$
4. I/Y $= 2, 4, 6, 8, 10, \ldots$
5. CPT N

Example 2.7

You are offered \$100 now plus \$100 in a year, or \$400 at a later time. If the two options are equivalent at $i = 12\%$ when would you receive the \$400?

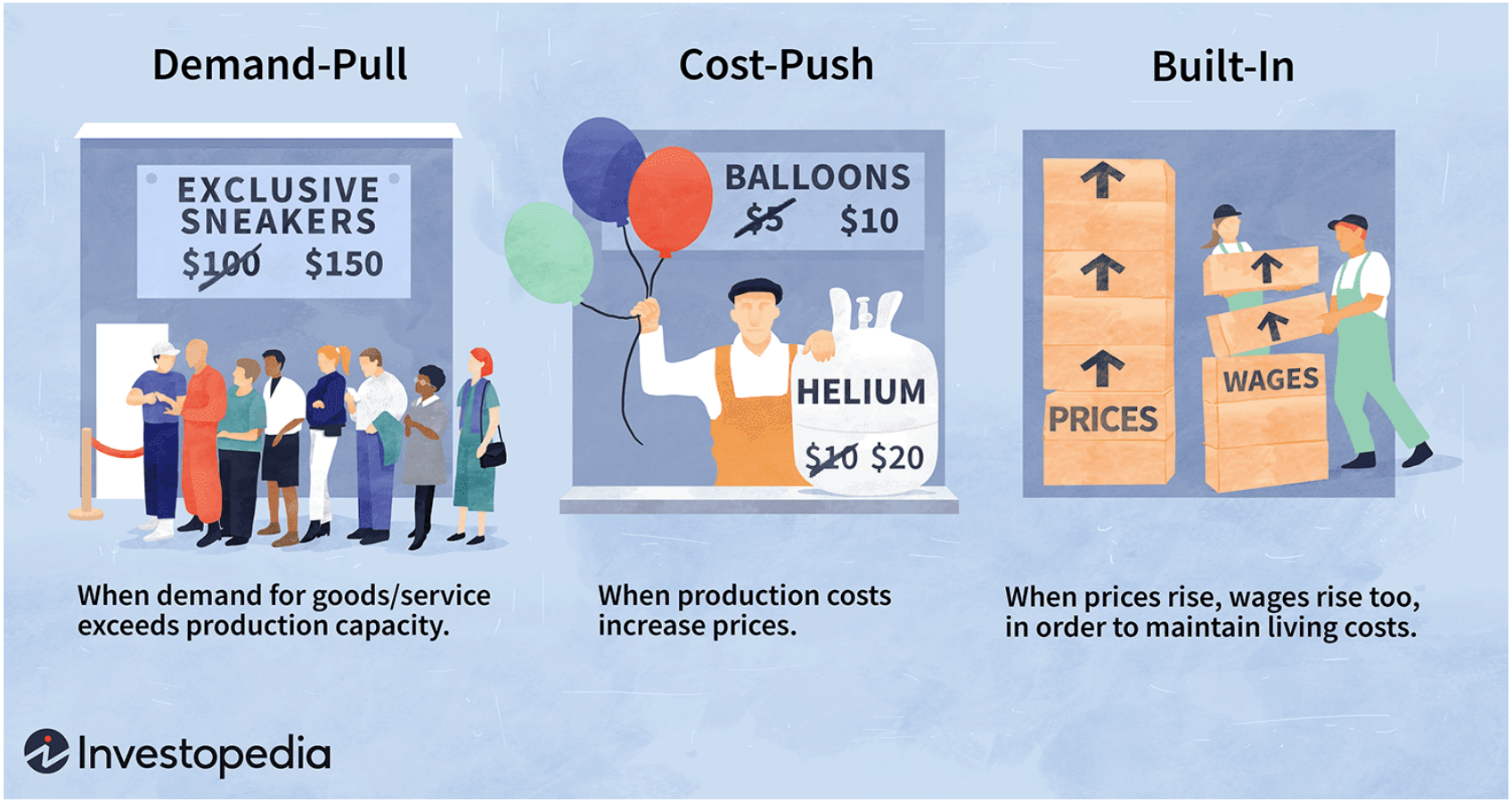
2.3 Inflation & real rates of interest

Inflation refers to the general increase in the price of goods over time.



Inflation erodes the purchasing power of money.

Why do we have inflation?



Inflation rate

The inflation rate is the (annualized) rate of change in the prices of consumer goods.

Price of milk (4L) in Dec 2021: \$5.07
Price of milk (4L) in Nov 2022: \$6.27
Price of milk (4L) in Nov 2024: \$6.63!

For specific goods, see Statistics Canada.

The Consumer Price Index (CPI)

- Inflation is measured by reference to an index that captures the average price of a specified basket of goods at different points in time.
- Inflation can be specific to a segment or sector of the economy.
- A widely used index is the Consumer Price Index (CPI).
- Statistics Canada constructs and estimates the Canadian CPI. See here to learn how.
- For information about the CPI in 2021, see here.

Hyperinflation

- Hyperinflation refers to a very rapid increase in the price of goods and services: over 50% per month (Cagan, 1956)
- Hanke and Krus (2013) compiled a list of 56 hyperinflation events from around the world.
- The highest level of inflation occurred in Hungary
  - monthly inflation rate (July 1946): $4.19 \times 10^{16}\%$!
  - equivalent daily inflation rate: 207%
  - time required for prices to double: 15 hours...

Hyperinflation





1 billion trillion =
1,000,000,000,000,000,000,000



Inflation and interest rates

Inflation counteracts investment growth.

Example 2.8: You have \$1000 today and inflation is 5% per year. What is the relative purchasing power of your \$1000 after one year (if not invested)?

You invest your \$1000 at $i = 12\%$. What is the relative purchasing power of the accumulated value at the end of the year?

The real interest rate

Example 2.8, continued:

What is the rate of interest earned during the year, after adjusting for inflation? 6.666%

We call this the **real/net/inflation-adjusted rate of interest** and denote it by i_{real}.

$$i_{real} = \frac{1+i}{1+r} - 1$$

Another "nominal" interest rate

The interest rate that has not been adjusted for inflation is called the **nominal interest rate**.

Note that "nominal" here means something different than in Section 1.7!

Example 2.9

Amy expects to receive four annual scholarship payments. The first payment (due now) is \$10,000. Each successive payment is adjusted for inflation so that in real terms the payments remain level. Inflation is 3% per year and the effective interest rate is 6% a year. Find the present value of Amy's scholarship.

Finding the present value

On your financial calculator

1. N = the number of compounding periods
2. I/Y (real interest rate per compounding period)
3. FV = future value (how much?)
4. PMT = periodic payments
5. CPT PV = − present value

Negative real rates of interest

When inflation (r) is higher than the nominal interest rate (i), the real rate of interest is negative. When this happens, there is an incentive to borrow.

Example 2.10: You borrow \$1000 for one year. Interest is charged at 2% and inflation is 4%. How much do you have to pay back at the end of the year

(a) in nominal terms?

(b) in real terms?

References

Cagan, P. (1956). 'The monetary dynamics of hyperinflation', in Friedman, M. (ed.) *Studies in the Quantity Theory of Money.* Chicago, IL: University of Chicago Press.

Hanke, S.H. and Krus, N. (2013) 'World hyperinflations', in Parker, R.E. and Whaples, R.M. (eds.) In *Routledge Handbook of Major Events in Economic History.* Routledge.